

EASWARI ENGINEERING COLLEGE
(Autonomous)
Bharathi Salai, Ramapuram, Chennai – 89

DEPARTMENT OF INFORMATION TECHNOLOGY

TOPIC : SMART FACTORY COLLABORATIVE

WORKSPACE MONITORING SYSTEM

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UNDER GUIDANCE OF

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ABSTRACT

1. **Smart factories** rely on collaborative workspaces shared by employees and machines every day.

2. **Traditional monitoring** depends on manual checks and disconnected tools, which is slow and inconsistent.

3. **This project proposes** an IoT and AI-based system to monitor workspaces automatically in real time.

4. **The system tracks** employee-machine interaction, equipment status, and workspace conditions continuously.

5. **The goal is** safer operations, higher productivity, and better coordination across the factory floor.

INTRODUCTION

1. **Modern factories** increasingly use shared spaces where employees and machines work side by side.

2. **This shift** has increased the need for real-time visibility into workspace activity and safety.

3. **Manual supervision** is slow, limited in coverage, and prone to human error.

4. **Smart factories need** connected sensors and intelligent systems that monitor continuously.

5. **This project** uses IoT and data analytics to build a real-time collaborative workspace monitoring system.

LITERATURE SURVEY

Author	Year	Description	Pros	Cons
Villani et al.	2018	Surveyed human-robot collaboration safety and interface design in industrial settings.	Established key safety and interaction guidelines.	Focused mainly on robot safety, less on full workspace monitoring.
Saleem et al.	2025	Reviewed external sensors (cameras, radars) for detecting humans in shared robot workspaces.	Comprehensive comparison of sensing technologies.	Sensor fusion adds system complexity and cost.
Maity et al.	2024	Proposed an IoT sensor network for real-time monitoring of machine temperature, vibration and status.	Enabled predictive maintenance and reduced downtime.	Requires reliable connectivity and regular sensor calibration.

EXISTING SYSTEM

Most factories currently rely on manual supervision and disconnected tools to track workspace activity.

MS

Manual Supervision

Supervisors physically check workspace activity and safety at intervals.

SS

Standalone Sensors

Individual sensors monitor specific machines but don't share data with each other.

CC

Basic CCTV Systems

Cameras record activity but need manual review, with no real-time alerts.

DISADVANTAGES

1. **Delayed Response –** Manual checks cannot catch issues as they happen.

2. **Limited Coverage –** Supervisors cannot monitor every workspace at once.

3. **Disconnected Data –** Standalone sensors don't share data for a complete picture.

4. **No Predictive Insight –** Existing tools react to problems instead of preventing them.

PROPOSED SYSTEM

RT

Real-Time Monitoring

IoT sensors track machine status, employee activity, and workspace conditions continuously.

SC

Smart Collaboration Engine

Combines sensor data, computer vision, and analytics to detect issues and coordinate tasks.

CI

Continuous Insights

Dashboards and alerts help supervisors act quickly and improve safety and productivity over time.

ADVANTAGES

1. **Higher Safety –** Detects hazards and unsafe conditions instantly.

2. **Real-Time Alerts –** Notifies teams immediately when issues occur.

3. **Better Coordination –** Improves communication between employees and machines.

4. **Data-Driven Decisions –** Supports reporting and continuous improvement.

SYSTEM REQUIREMENTS

HW Hardware Specifications

Processor
2.5 GHz or above

RAM
8 GB (Minimum)

Storage
100 GB

Input Device
Standard Keyboard & Mouse

Monitor
1920 × 1080 (Minimum)

Sensors
IoT Motion, Temperature & Proximity Sensors

SW Software Specifications

Operating System
Windows 10 or Above

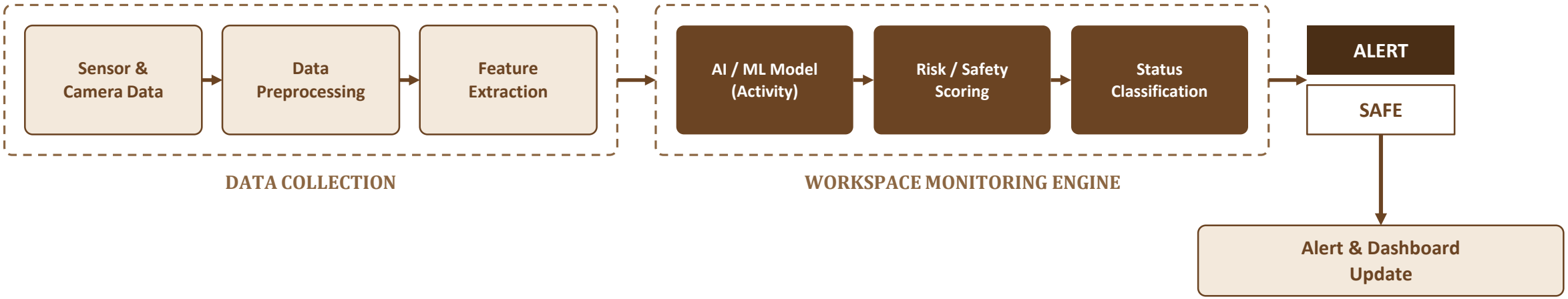
Frontend
HTML, CSS, JavaScript

Backend
Node.js, Express.js

Database
MongoDB

Tools
VS Code, GitHub, MQTT Broker

SYSTEM ARCHITECTURE



REFERENCES

1. Villani, V.; Pini, F.; Leali, F.; Secchi, C. Survey on human-robot collaboration in industrial settings: Safety, intuitive interfaces and applications. Mechatronics, 2018.

2. Saleem, Z.; Gustafsson, F.; Furey, E.; McAfee, M.; Huq, S. A review of external sensors for human detection in a human robot collaborative environment. Journal of Intelligent Manufacturing, 2025, 36(4), 2255-2279.

3. Li, W.; Hu, Y.; Zhou, Y.; Pham, D.T. Safe human-robot collaboration for industrial settings: A survey. Journal of Intelligent Manufacturing, 2024, 35(5), 2235-2261.

4. Maity, A.K.; Darshan; Chauhan, K. IoT-Based Industrial Equipment Monitoring System: Revolutionizing Maintenance through Smart Data Analytics. SSRN, 2024.

5. International Organization for Standardization. ISO/TS 15066:2016 Robots and Robotic Devices — Collaborative Robots, 2016.



THANK YOU!

Smart Factory Collaborative Workspace Monitoring System